ASHWIN SATISH

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EDUCATION

Northeastern University - Khoury College of Computer Science **Sept 2024 - Dec 2026 | Boston, MA**
Master of Science, Data Science **GPA**: 3.83/4.0
Machine Learning, NLP, MLOps, Algorithms, Data Mining, Database Management, Capstone

Christ University **June 2020 - May 2023 | Bangalore, India**
Bachelor of Science, Computer Science **GPA**: 3.53/4.0

EXPERIENCE

Institute for Experiential AI, Northeastern University | Machine Learning Developer Intern **Jan 2026 - Present | Boston, MA**

- Engineered an LLM pipeline, mining **35M** records, finding **278K**+ publication-trial links, transforming **13K**+ papers into structured data via strict **JSON Schema** outputs, with every field traced to its source sentence.
- Designed & shipped a clinical-trial data model in Neo4j (**115K+** nodes, **116K+** relationships), exposing it via REST APIs & a production Flask app.
- Architected a custom entity-normalization pipeline mapping **45K+** clinical-trial adverse events to standard medical ontologies, raising coverage from **75.1%** to **85.4%** (**+4.6K** events) with zero regressions.
- Benchmarked LLMs on AWS Bedrock for clinical-data extraction across cost, recall, & token efficiency, cutting per-paper inference cost **~65%** while retaining **90% recall** against the baseline model.

Northeastern University | Machine Learning Assistant **Jan 2025 - Dec 2025 | Boston, MA**

- Boosted true positives from **2% to 77%** (maintaining zero false positives**)** for Lyme cell detection via 7-model CNN ensemble.
- Built a **Python** image preprocessing pipeline for **30,000+** MATLAB cell images, saving the biology team weeks of manual work.
- Improved ensemble accuracy by **13%** via augmentation & L2/dropout regularization, validated with k-fold cross-validation.
- Mentored & conducted office hours as a Teaching Assistant for **300+ students** in Machine Learning, statistics & **NumPy** demos.

CarerAI | Data Science Intern **Jul 2023 - May 2024 | Gurgaon, India**

- Prevented **2,000+ unnecessary procedures** for customers by uncovering **500+** surgery avoidance cases through k-Nearest Neighbors (KNN) analysis of patient records against historical data.
- Leveraged EHR for a **risk-score**, improving classification accuracy by **26.4%**, increasing confidence in surgery avoidance cases.
- Improved KNN's score by **20%** using DBSCAN to remove record outliers, reducing false positives in surgery recommendations.

SKILLS

- **Languages**: Python, SQL, R, C, Java, TypeScript, JavaScript
- **Libraries & Frameworks**: Scikit-learn, TensorFlow, PyTorch, Keras, Hugging Face, OpenCV, Pandas, NumPy, PySpark
- **Cloud & Development**: AWS, GCP, Azure, PostgreSQL, Snowflake, Docker, Neo4j, Bedrock, Lambda, Kubeflow
- **LLM & GenAI Tooling**: Transformers, BERT, LoRA, AutoML, T5, LangChain, LangGraph, Ollama, PydanticAI, MLflow

PROJECTS

AI-Powered Pokemon Battle Strategy Engine | *Ollama, Recommendation System, Sentence-Transformers, Streamlit* | **GitHub**

- Built & dockerized hybrid **Multi-Agent System (MAS)** integrating Llama 3.2 with deterministic backend for strategy analysis & a detailed recommendation system for optimal synergies by ingesting **1M+** aggregated usage **stats** via Python **ETL** pipeline.
- Engineered a RAG pipeline via ChromaDB & Sentence-Transformers, indexing **40,000+** strategy docs to ground LLM inference.

Pantry & Recipe Assistant | ***2nd Place, MLOps Expo Google*** **|** *LoRA, PostgreSQL, Airflow, Docker, GCP, Ollama* **| GitHub**

- Fine-tuned, evaluated & served Llama 3.2 3B on GCP via CI/CD for personalized inventory recipes, cutting dietary violations by 50% through LoRA adaptation.
- Collaborated on **DPO feedback** with Teacher model (Llama 3 70B) for preference labelling, VLM for OCR & Veo 3.1 for video.

Airlytics for Fitbit: Analysis, Forecast & Health Coach | *FastAPI, SQLAlchemy, SQLite, Groq, Scikit-learn, Pandas* **| GitHub**

- Built personalized readiness-forecasting models & dashboards for Fitbit Air's data, beating a naive baseline by **16%**.
- Built LLM health coach that provides health insights on retrieved data, validated by scoring tool-call accuracy & grounding rate.

Casino Player Retention: A/B Testing & Analytics | *A/B Testing, SQLite, Bayesian Inference, SciPy, Streamlit* **| GitHub**

- Designed a 3-arm A/B test (frequentist + Bayesian) on 15K simulated players, detecting a 14% relative lift in 30-day retention.
- Built a SQLite analytics layer (recursive CTEs, window functions) & Streamlit dashboard for retention, LTV & results.